

DSAT MINOR EXAM

Name :- Manvendra pratap Singh

Roll No: G24 AIT 2163

Solution 3

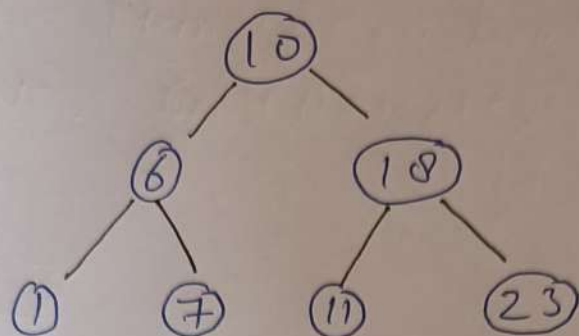
To construct the BST in different heights using given set $\{1, 6, 7, 10, 11, 18, 23\}$, we need to follow some rules,

- (i) height of BST is no. of edges on the longest path from root to leaf
- (ii) Balance BST's will have smaller heights
- (iii) Unbalanced BST :- may have greater height.

BST for Different Heights

Conti...

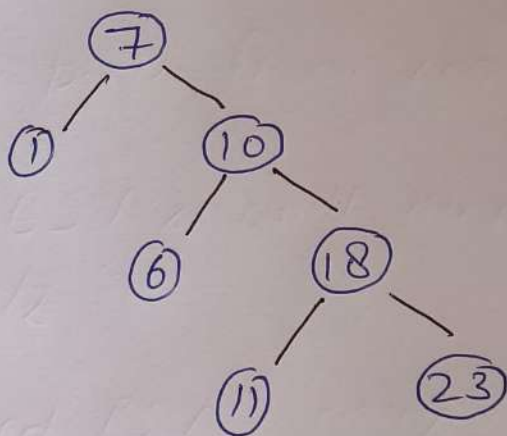
BST's height 2 (Balanced BST)



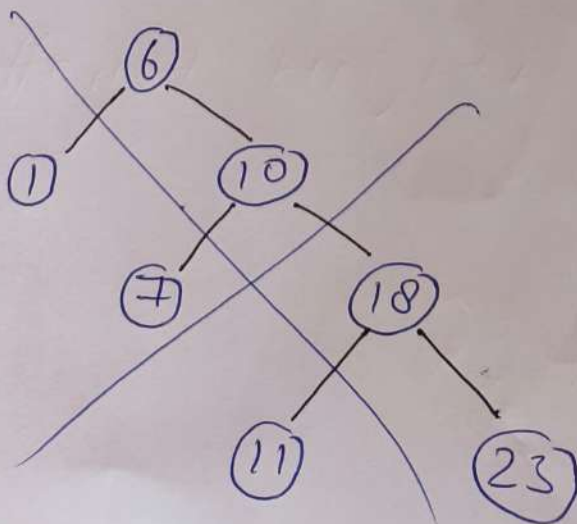
$10 \rightarrow 1$, $10 \rightarrow 7$, $10 \rightarrow 11$, $10 \rightarrow 23$

height = 2

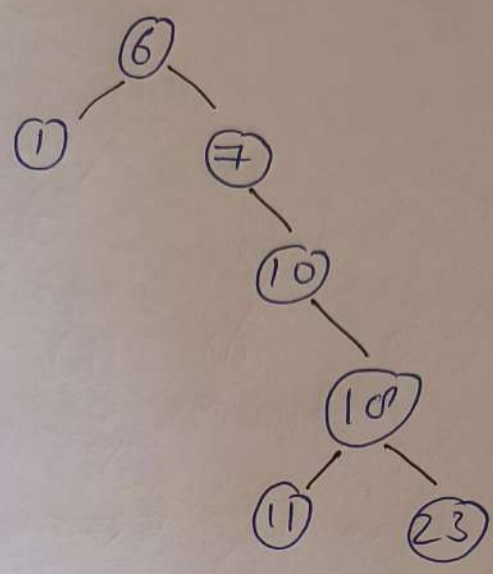
Height (3)



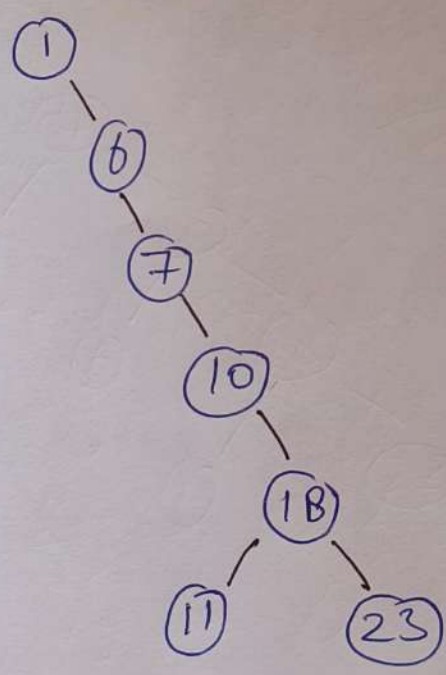
~~Height (4)~~



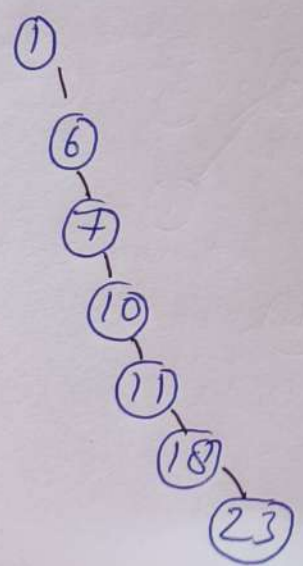
Height ~~4~~ (4)



Height ~~4~~ (5)



Height (6)



(Skewed Tree.)

1 → 23

Height = 6

Solution 6

To, Achieve the required ordering
 $A[0] < A[1] > A[2] < A[3] > A[4] \dots$

We follow the Zigzag - Based $O(n)$

To achieve the result we can say that

① Even indices (0, 2, 4, ...) $\Rightarrow$ smaller value than the adj. odd indices

② odd indices (1, 3, 5, ...) $\Rightarrow$ greater value than neighbors.

Pseudocode

func re-ordering (A)

$n = \text{length}(A)$

for ($i = 0$ to $n - 2$)

if ($i \% 2$) // even

if ($A[i] > A[i+1]$)

swap ($A[i], A[i+1]$)

else

// odd

if ($A[i] < A[i+1]$)

5

swap($A[i]$, $A[i+1]$)

end the loop

return A

Explanation:

We traverse the array once & swap the elements in-place where the condⁿ is not fulfilled. So complexity $O(n)$

Step by Step

① Traverse the array from 0 to length-2

② Check the condⁿ at each index i

~~if i is even swap both~~

→ if i is even & $A[i] > A[i+1]$
swap both $A[i]$ & $A[i+1]$

→ if i is odd & $A[i] < A[i+1]$
swap both $A[i]$ & $A[i+1]$

contin. till length

Given example

3, 1, 0, 4, 8, 6, 10

Index	Cond ⁿ	Swap	Updated Array
i=0	3 > 1	Swap	1, 3, 0, 4, 8, 6, 10
i=1	3 > 0	No-Swap	"
i=2	0 < 4	No Swap	1, 3, 0, 4, 8, 6, 10
i=3	4 < 8	Swap	1, 3, 0, 8, 4, 6, 10
i=4	4 < 6	No-Swap	"
i=5	6 < 10	Swap	1, 3, 0, 8, 4, 10, 6

Final Solⁿ

[1, 3, 0, 8, 4, 10, 6]

Time Complexity Analysis

~~Best case $\rightarrow$ all sorted $O(1)$~~
~~Worst case $\rightarrow$ all not in condⁿ $O(n)$~~

① $O(n)$, without extra space

PseudocodeSolution 5Pseudocode

Hash-process (arr, n)

freq. = Empty-hash-table

for i from 0 to n-1

if arr[i] in freq

freq[arr[i]] = freq[arr[i]] + 1

else

freq[arr[i]] = 1

prefix = Empty-hash-table

keys = Sorted-keys(freq)

sum = 0

for key in keys

sum = sum + freq[key]

prefix[key] = sum

Return prefix

Query (prefix, a, b)

// Prefix sum hash table

if b not in prefix

Return 0

if a == 0

Return prefix[b]

if a-1 not in prefix

Return prefix[b]

Return prefix[b] - prefix[a-1]

Complexity

→ preprocessing (Hash-process) - $O(n \log n)$

→ querying - $O(1)$

So $O(n \log n)$

Space complexity - $O(n)$

Solution 2

Primary hash function

$$h_1(K) = K \bmod 11$$

$$h_2(K) = 1 + (K \bmod (m-1))$$
$$= 1 + (K \bmod 10)$$

$$h(K, i) = (h_1(K) + i h_2(K)) \bmod 11$$

Insert 34

$$h_1(34) = 34 \bmod 11 = 1$$

$$h_2(34) = 1 + (34 \bmod 10) = 1 + 4 = 5$$

Insert 34 at 1

insert 21

$$h_1(21) = 21 \bmod 11 = 10$$

$$h_2(21) = 1 + (21 \bmod 10) = 1 + 1 = 2$$

insert 21 at 10

insert 11

$$h_1(11) = 0$$

$$h_2(11) = 1 + 1 = 2$$

insert 11 at 0

insert 45

$$h_1(45) = 1 \text{ collision}$$

$$h_2(45) = 1 + 5 = 6$$

$$(1 + 1 \times 6) \% 11 = 7 \% 11 = 7$$

Insert 45 at 7

Insert 8

$$h_1(8) = 8$$

$$h_2(8) = 1 + 8 = 9$$

insert 8 at 8

Insert 7

$$h_1(7) = 7 \% 11 = 7 \text{ // collision}$$

$$h_2(7) = 1 + (7 \% 10) = 1 + 7 = 8$$

$$(7 + 1 \times 8) \% 11 = 15 \% 11 = 4$$

insert 7 at 4

Insert 58

$$h_1(58) = 58 \% 11 = 3$$

insert 99

$$h_1(99) = 99 \% 11 = 0 \text{ // collision}$$

(11)

$$h_2(99) = 1 + (99 \times 10) = 1 + 9 = 10$$

$$\Rightarrow (0 + 1 \times 10) \times 11 = 10 \times 11 = 10$$

11 collision 10

$$\Rightarrow (0 + 2 \times 10) \times 11 = 20 \times 11 = 9$$

Insert 99 at (9)

Final Hash table

Index	Key
0	11
1	34
2	-
3	58
4	7
5	-
6	-
7	45
8	8
9	99
10	21

Solution (1)

- a) True
- b) True
- c) false
- d) false
- e) True.

Solution 4

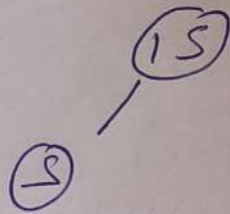
a) No, inserting n & then y is not always the same as inserting y & then n in a BST.

Example

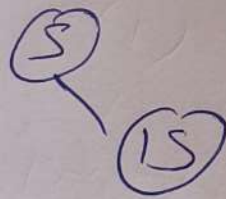
Suppose an initial empty BST

take $n = 15$ & $y = 5$

BST - n then y



y then n



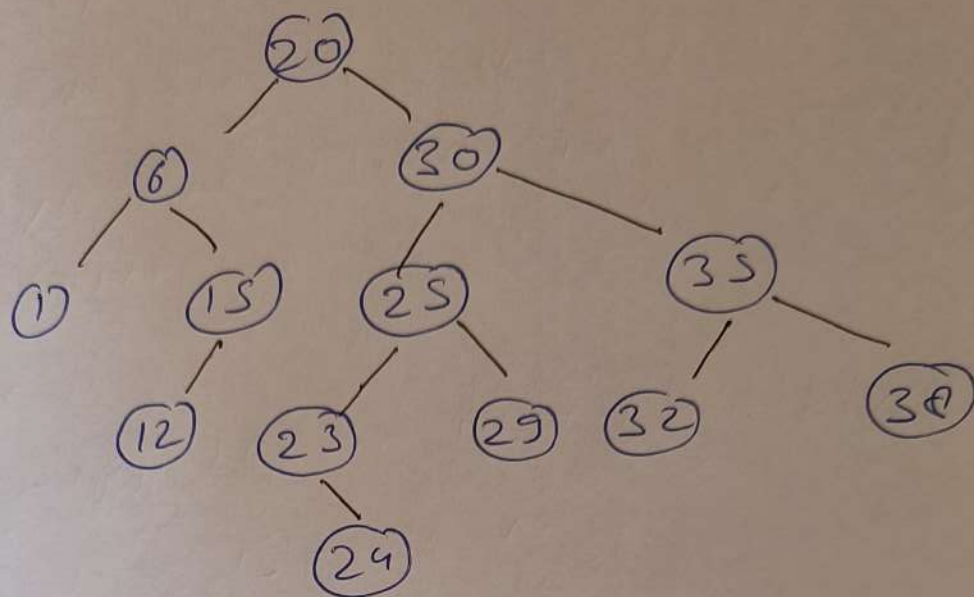
Conclusion

Inserting n & then y don't always give the same BST as inserting y then n .

⇒ BST insertion depends on the order of insertion & each new node is placed a/c to its relation with the existing nodes.

∴ the statement is FALSE.

Solution
4 b)



delete 20

Here, 20 has both child & sub-child
So we need to find the successor of
20 i.e. 23 & replace 20 → 23
& re-link 24 → 23

Now BST is

